

Statistics

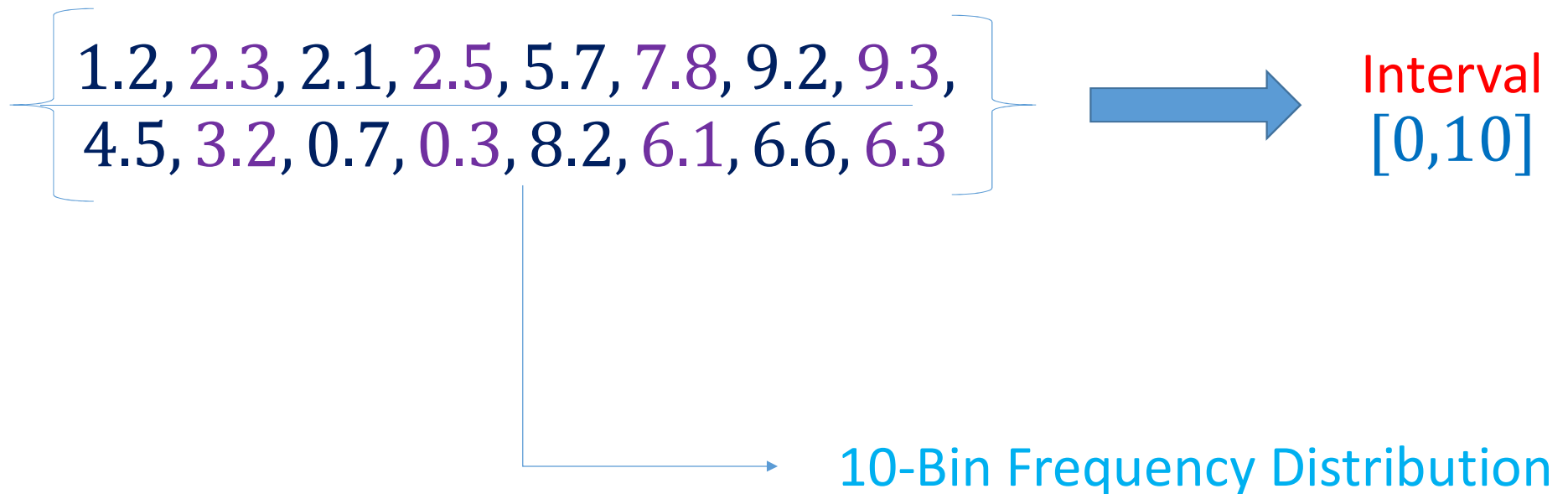


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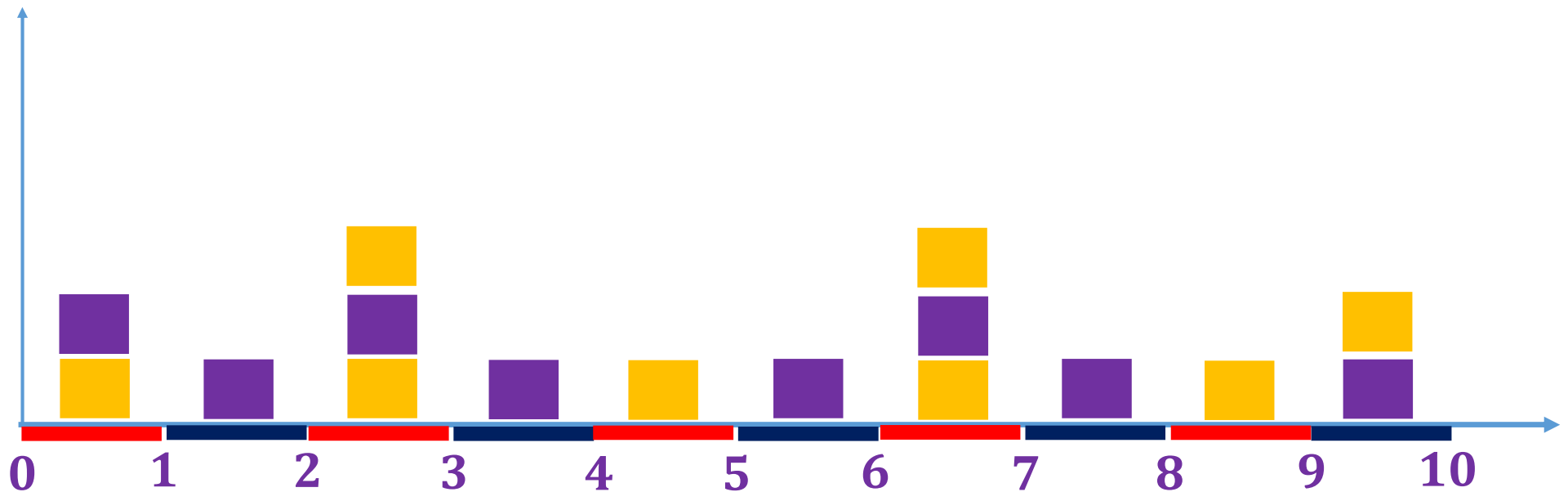
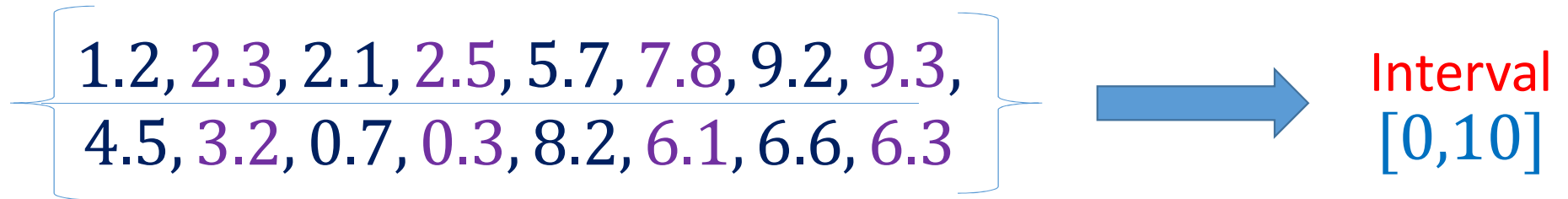
Overview

- Frequency Distribution
- Sum Normalized Frequency Distribution
- Measures of Central Tendency
- Standard Deviation & Variance
- Covariance Matrix
- Mahalanobis Distance
- Coordinate Transformation
- Interpreting Covariance Matrix

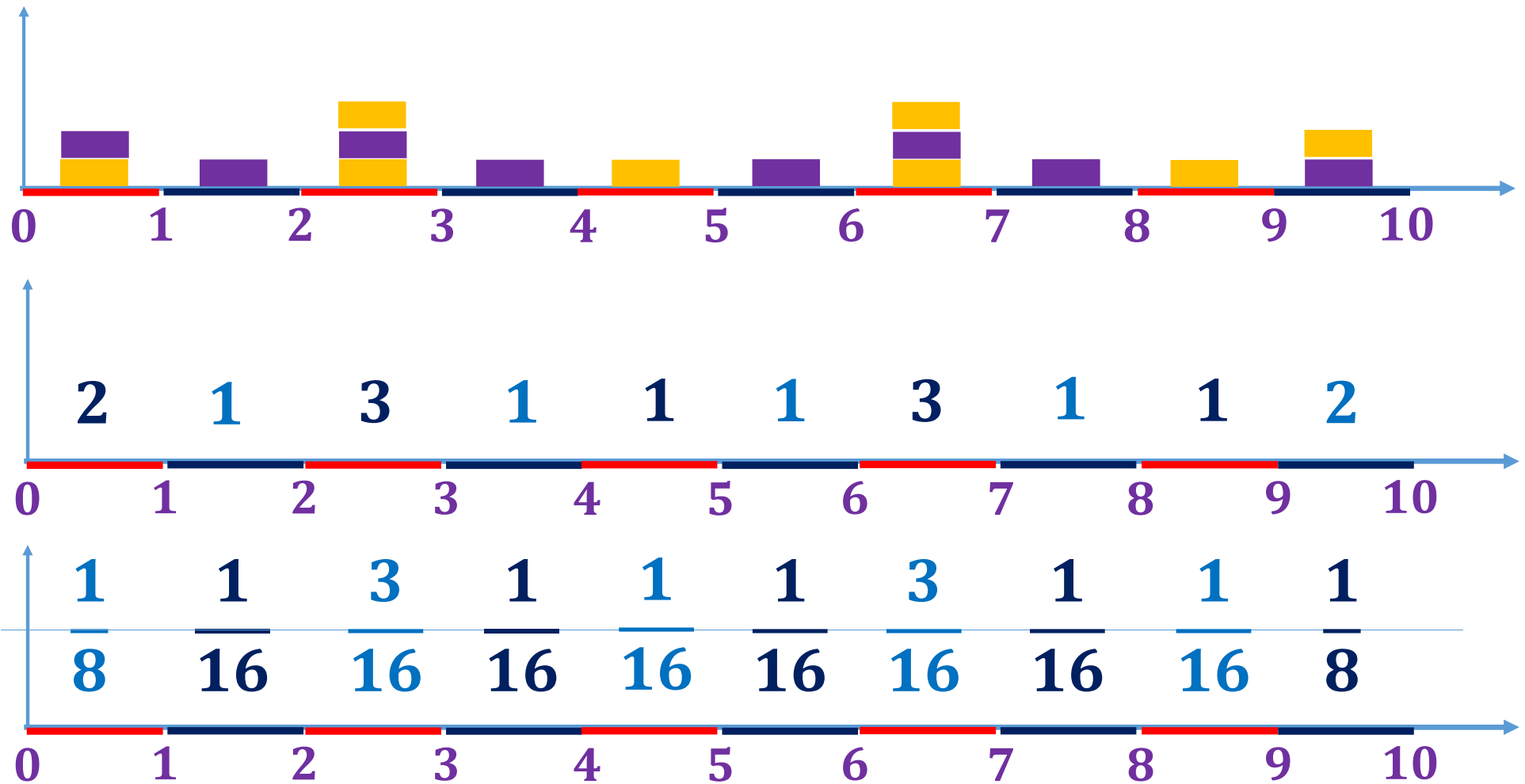
Frequency Distribution & Histogram



Frequency Distribution & Histogram



Sum Normalized Histogram

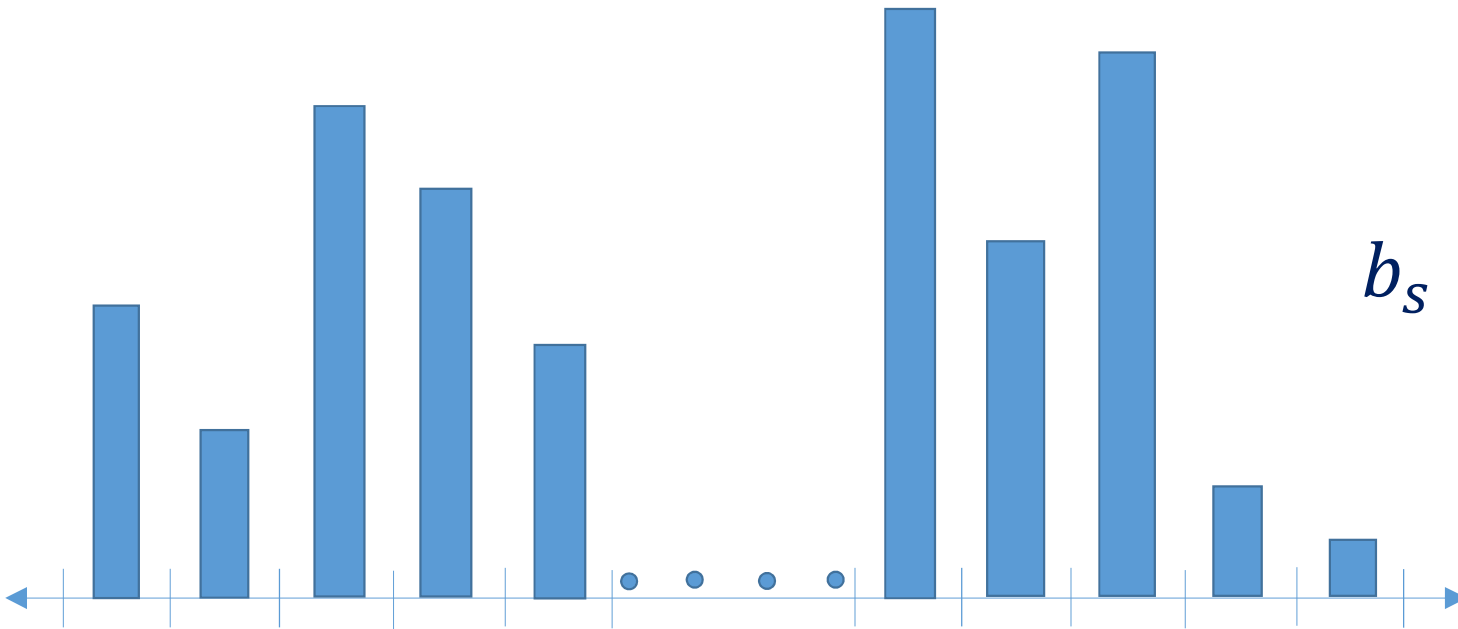


Frequency Distribution

$$\mathbf{S} = \{x_1, \dots x_i, \dots x_n\}$$

$\min(\mathbf{S}) = x_{\min}$

$\max(\mathbf{S}) = x_{\max}$



$$b_s = \frac{x_{\max} - x_{\min}}{m}$$

Normalized Frequency Distribution: Pseudo-code

- INPUT: $\mathbf{S} = \{x_1, \dots x_i, \dots x_n\}$
- Intervals: $x_{max} = \max(\mathbf{S}); x_{min} = \min(\mathbf{S})$
- Bin: $b_s = \frac{x_{max} - x_{min}}{m}$
- INITIALIZE: $\mathbf{H}[j] = 0; j = 1, \dots m$
- FOR $i = 1 \rightarrow n$
 1. Get Bin Index: $b_i = \left\lfloor \frac{x_i - x_{min}}{b_s} \right\rfloor + 1$
 2. UPDATE: $\mathbf{H}[b_i] = \mathbf{H}[b_i] + \frac{1}{n}$
- END FOR

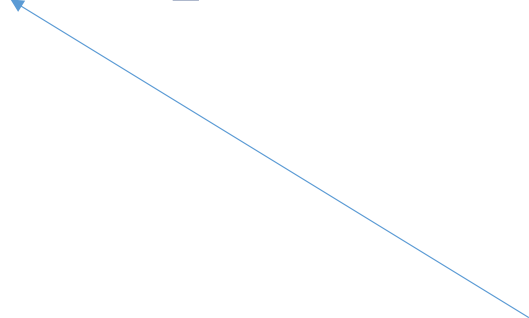
Kronecker Delta Function

$$\delta(i, j) = \delta[i - j] = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Normalized Frequency Distribution: Mathematical Expression

$$H[j] = \frac{1}{n} \sum_{i=1}^n \delta \left[\left\lfloor \frac{x_i - x_{min}}{b_s} \right\rfloor + 1 - j \right]$$


$$j = 1, \dots, m$$


$$b_s = \frac{x_{max} - x_{min}}{m}$$

Normalized Weighted Frequency Distribution

$$H[j] = \frac{1}{W} \sum_{i=1}^n w(x_i) \delta \left[\left\lfloor \frac{x_i - x_{min}}{b_s} \right\rfloor + 1 - j \right]$$

$j = 1, \dots, m$

$$W = \sum_{i=1}^n w(x_i)$$
$$b_s = \frac{x_{max} - x_{min}}{m}$$

Measures of Central Tendency

- Arithmetic Mean
- Geometric Mean
- Harmonic Mean
- Weighted Mean

- Median
- Weighted Median

- Mode
- Weighted Mode

Arithmetic Mean (μ_A)

$$\mu_A = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + \cdots + x_i + \cdots x_n}{n}$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\}$$

$$\mu_A = \frac{1.2 + 2.5 + 4.5 + 6.6 + 8.2}{5} = \frac{23}{5} = 4.6$$

Geometric Mean (μ_G)

$$\mu_G = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} = (x_1 \times \cdots \times x_i \times \cdots \times x_n)^{\frac{1}{n}}$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\}$$

$$\mu_G = (1.2 \times 2.5 \times 4.5 \times 6.6 \times 8.2)^{\frac{1}{5}} = 3.7389$$

Harmonic Mean (μ_H)

$$\mu_H = \left(\frac{1}{n} \sum_{i=1}^n x_i^{-1} \right)^{-1} = \frac{n}{\frac{1}{x_1} + \dots + \frac{1}{x_i} + \dots + \frac{1}{x_n}}$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\}$$

$$\mu_H = \frac{5}{\frac{1}{1.2} + \frac{1}{2.5} + \frac{1}{4.5} + \frac{1}{6.6} + \frac{1}{8.2}} = 2.8918$$

$$\mu_A \geq \mu_G \geq \mu_H$$

Weighted Arithmetic Mean (μ_A^w)

$$\mu_A^w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} = \frac{w_1 x_1 + \cdots + w_i x_i + \cdots + w_n x_n}{w_1 + \cdots + w_i + \cdots + w_n}$$

$$W = \{0.1, 0.5, 0.2, 0.9, 0.5\}$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\}$$

$$\mu_A = \frac{0.1 \times 1.2 + 0.5 \times 2.5 + 0.2 \times 4.5 + 0.9 \times 6.6 + 0.5 \times 8.2}{0.1 + 0.5 + 0.2 + 0.9 + 0.5} = \frac{12.21}{2.2} = 5.596$$

Median (ρ)

Median Divides Data Distribution into Two Equal Halves

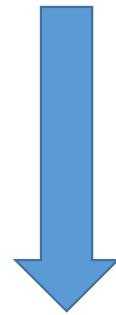
{ 1.2, 2.3, 2.1, 2.5, 5.7, 7.8, 9.2, 9.3,
4.5, 3.2, 0.7, 0.3, 8.2, 6.1, 6.6, 6.3 }

→ $\rho = ?$

Median (ρ)

Median Divides Data Distribution into Two Equal Halves

{ 1.2, 2.3, 2.1, 2.5, 5.7, 7.8, 9.2, 9.3,
4.5, 3.2, 0.7, 0.3, 8.2, 6.1, 6.6, 6.3 }



Sort in
Ascending
Order

{ 0.3, 0.7, 1.2, 2.1, 2.3, 2.5, 3.2, 4.5,
5.7, 6.1, 6.3, 6.6, 7.8, 8.2, 9.2, 9.3 }

$$\rho = \frac{4.5 + 5.7}{2} = 5.1$$

Weighted Median (ρ_w)

Median Divides Data Distribution into Two Equal Halves

$$W = \{w_1, \dots, w_i, \dots, w_n\}$$

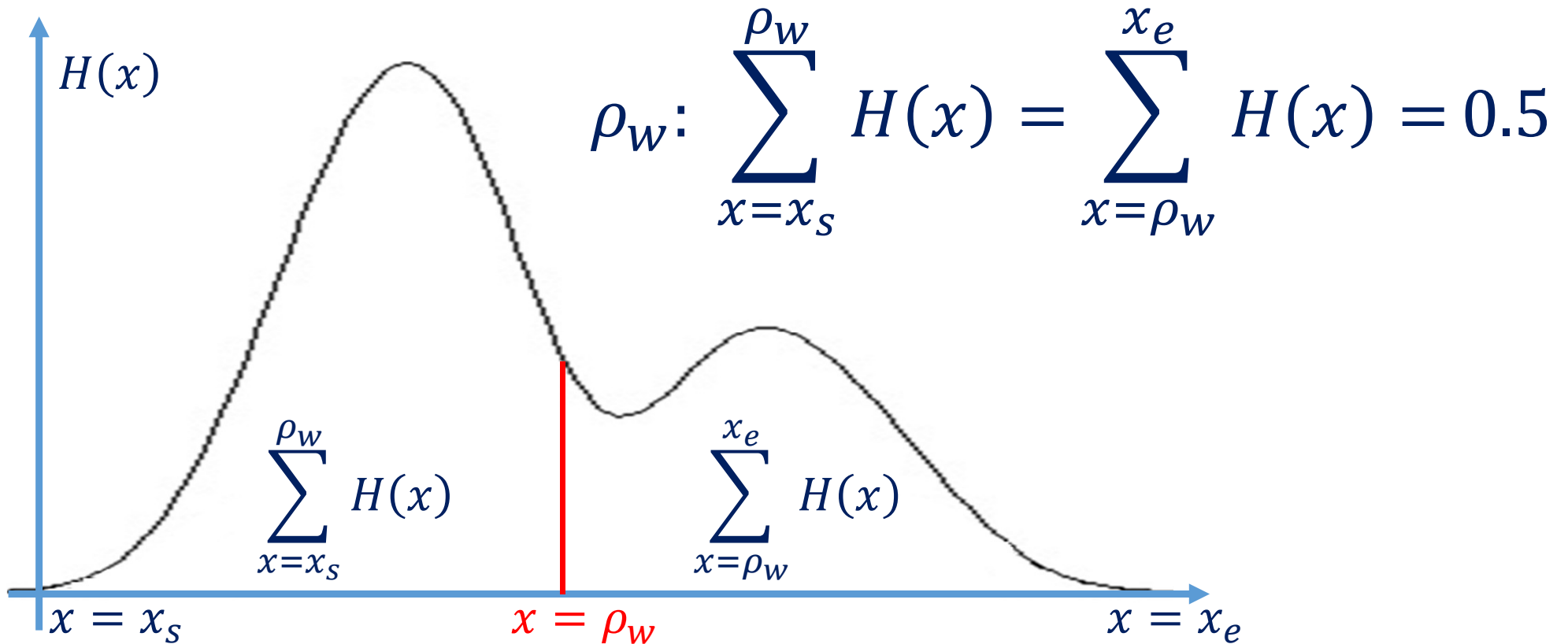
$$S = \{x_1, \dots, x_i, \dots, x_n\}$$

Construct Normalized Frequency Distribution $H(x)$

$$\rho_w: \sum_{x=x_s}^{\rho_w} H(x) = \sum_{x=\rho_w}^{x_e} H(x) = 0.5$$

Weighted Median (ρ_w)

Median Divides Data Distribution into Two Equal Halves



Weighted Median (ρ_w)

x	4	-6	0	8	-2	10	-4	2	-8	6
w	4	2	3	2	2	4	3	4	3	3

$$\rho_w = ?$$

Weighted Median (ρ_w)

x	4	-6	0	8	-2	10	-4	2	-8	6
w	4	2	3	2	2	4	3	4	3	3



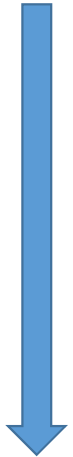
Variates Sorted in Ascending Order

x	-8	-6	-4	-2	0	2	4	6	8	10
w	3	2	3	2	3	4	4	3	2	4

Weighted Median (ρ_w)

x	-8	-6	-4	-2	0	2	4	6	8	10
w	3	2	3	2	3	4	4	3	2	4

Sum Normalization of Weights


$$p(i) = \frac{w(i)}{\sum_{j=1}^n w(j)}$$

x	-8	-6	-4	-2	0	2	4	6	8	10
p	0.10	0.067	0.10	0.067	0.10	0.133	0.133	0.10	0.067	0.133

Weighted Median (ρ_w)

x	-8	-6	-4	-2	0	2	4	6	8	10
p	0.10	0.067	0.10	0.067	0.10	0.133	0.133	0.10	0.067	0.133



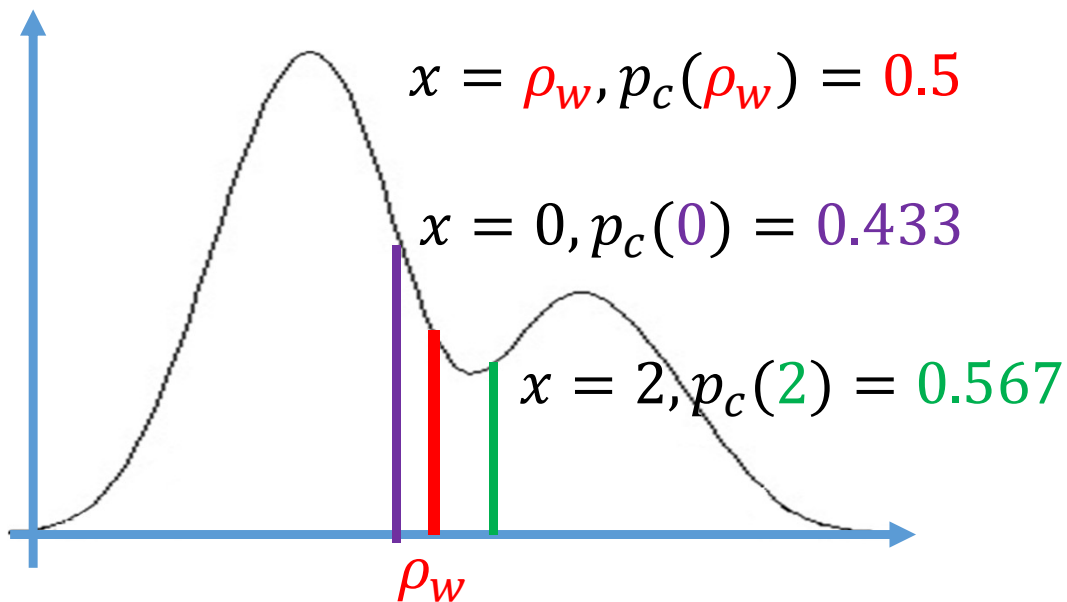
Cumulative Probability Computation

$$p_c(i) = p_c(i - 1) + p(i)$$

x	-8	-6	-4	-2	0	2	4	6	8	10
p_c	0.10	0.167	0.267	0.333	0.433	0.567	0.7	0.8	0.867	1.0

Weighted Median (ρ_w)

x	-8	-6	-4	-2	0	2	4	6	8	10
p_c	0.10	0.167	0.267	0.333	0.433	0.567	0.7	0.8	0.867	1.0



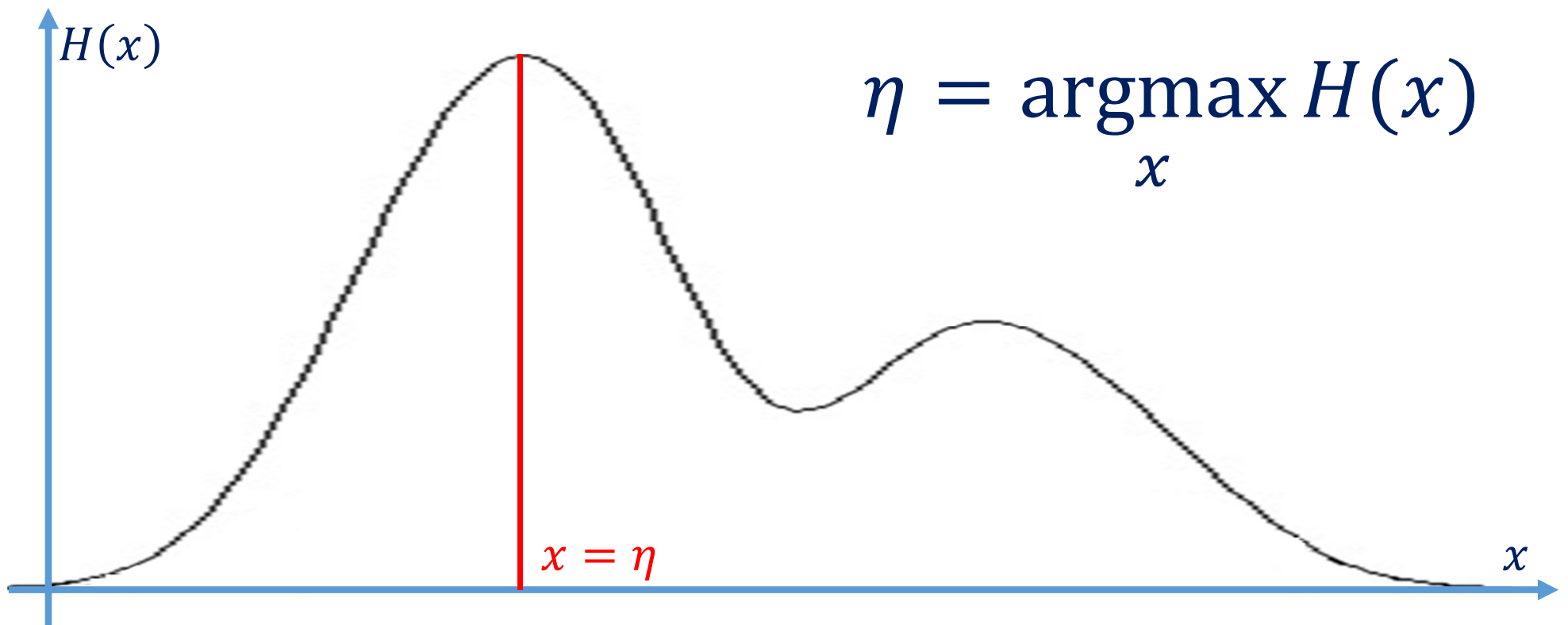
$$\frac{0.567 - 0.433}{2 - 0} = \frac{0.5 - 0.433}{\rho_w - 0}$$

$$0.067 = \frac{0.067}{\rho_w}$$

$$\rho_w = 1$$

Weighted Mode (η)

Peak of the Distribution is obtained at the Mode



Measures of Dispersion

- Range
- Average Absolute Deviation
- Median Absolute Deviation
- Standard Deviation

Range (***R***)

$$R = \max(\mathbf{X}) - \min(\mathbf{X})$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\}$$

$$R = 8.2 - 1.2 = 7$$

Average Absolute Deviation (AAD)

$$AAD = \frac{1}{n} \sum_{i=1}^n |x_i - \mu_A|$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\} \quad \mu_A = 4.6$$

$$MAD = \frac{|1.2-4.6|+|2.5-4.6|+|4.5-4.6|+|6.6-4.6|+|8.2-4.6|}{5} = 2.24$$

Median Absolute Deviation (MAD)

$$MAD = \text{Median}\{|x_i - \rho|; i = 1, \dots, n\}$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\} \quad \rho = 4.5$$

$$MAD = \text{Median}\{|1.2 - 4.5|, |2.5 - 4.5|, |4.5 - 4.5|, |6.6 - 4.5|, |8.2 - 4.5|\}$$

$$MAD = 2.1$$

Standard Deviation (σ) & Variance (σ^2)

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_A)^2 = \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right) - \mu_A^2$$

$$S = \{1.2, 2.5, 4.5, 6.6, 8.2\} \quad \mu_A = 4.6$$

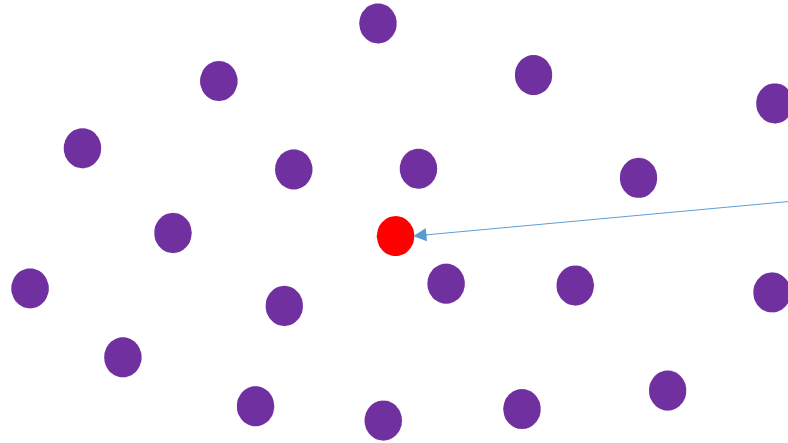
$$\sigma^2 = \frac{(1.2 - 4.6)^2 + (2.5 - 4.6)^2 + (4.5 - 4.6)^2 + (6.6 - 4.6)^2 + (8.2 - 4.6)^2}{5} = 6.588$$

Scalar Dispersion d_{mv} for Multivariate Data

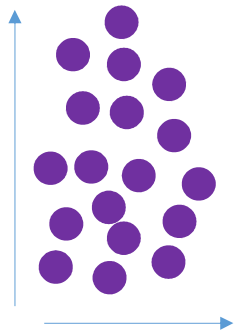
$$\mathbf{x}_i \in \mathbb{R}^d$$

$$\boldsymbol{\mu} \in \mathbb{R}^d$$

$$d_{mv} \in \mathbb{R}^1$$



$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$$



Different Dispersions
in Different Directions

$$d_{mv}^2 = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \boldsymbol{\mu}_A\|_2^2$$

d_{mv} Does Not Capture Directionality of Dispersion

Covariance $Cov(X, Y)$

$$X = \{x_1, \dots, x_i, \dots, x_n\}$$



$$\mu_X$$

$$Y = \{y_1, \dots, y_i, \dots, y_n\}$$



$$\mu_Y$$

$$Cov(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y) = \left(\frac{1}{n} \sum_{i=1}^n x_i y_i \right) - \mu_x \mu_y$$

$$Cov(X, Y) = Cov(Y, X)$$

Correlation $\rho(X, Y)$

$$X = \{x_1, \dots, x_i, \dots, x_n\}$$



$$\mu_X$$

$$Y = \{y_1, \dots, y_i, \dots, y_n\}$$



$$\mu_Y$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

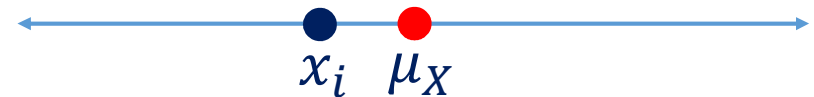
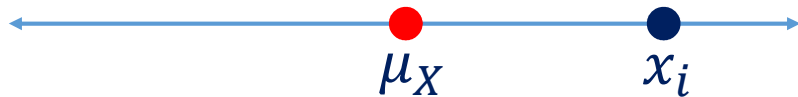
$$\rho(X, Y) = \rho(Y, X)$$

Pearson Product-Moment Correlation Coefficient

Positively Correlated

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

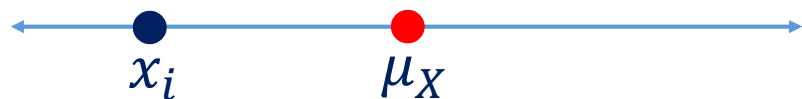


On an Average (x_i, y_i) lie on same side of (μ_X, μ_Y)

Negatively Correlated

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

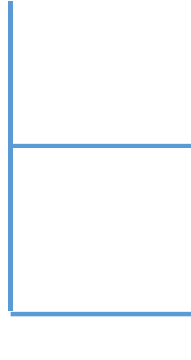
$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$



On an Average (x_i, y_i) lie on opposite sides of (μ_X, μ_Y)

Covariance Matrix ($\mathbf{C}$)

$$\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_i, \dots, \mathbf{x}_n\} \quad \mathbf{x} \in \mathbb{R}^2$$


$$\begin{aligned} \mathbf{X}_1 &= \{\mathbf{x}_1[1], \dots, \mathbf{x}_i[1], \dots, \mathbf{x}_n[1]\} \\ \mathbf{X}_2 &= \{\mathbf{x}_1[2], \dots, \mathbf{x}_i[2], \dots, \mathbf{x}_n[2]\} \end{aligned}$$

$$\mathbf{C} = \begin{bmatrix} Var(\mathbf{X}_1) & Cov(\mathbf{X}_1, \mathbf{X}_2) \\ Cov(\mathbf{X}_2, \mathbf{X}_1) & Var(\mathbf{X}_2) \end{bmatrix}$$

Covariance Matrix ($\mathbf{C}$)

$$\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_i, \dots, \mathbf{x}_n\}$$

$$\mathbf{x} \in \mathbb{R}^d$$

$$\mathbf{X}_r = \{\mathbf{x}_1[r], \dots, \mathbf{x}_i[r], \dots, \mathbf{x}_n[r]\}$$

$$r = 1, \dots, d$$

$$\mathbf{C} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T$$

$$\mathbf{C} = \mathbf{C}^T$$

$$C[j][k] = C[k][j] = \text{Cov}(\mathbf{X}_j, \mathbf{X}_k) = \text{Cov}(\mathbf{X}_k, \mathbf{X}_j)$$

Covariance Matrix ($\mathbf{C}$)

$$\mathbf{X} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \end{bmatrix} \right\}$$



$$\mathbf{C} = ?$$

Covariance Matrix ($\mathbf{C}$)

$$\mathbf{X} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \end{bmatrix} \right\}$$

$$\mathbf{X}_1 = \{1, 0, 3\}$$

$$\mu_1 = \frac{1 + 0 + 3}{3} = 1.33$$

$$\sigma_1^2 = \left(\frac{1^2 + 0 + 3^2}{3} \right) - (1.33)^2 = 2.56$$

$$\mathbf{X}_2 = \{-1, 2, -4\}$$

$$\mu_2 = \frac{-1 + 2 - 4}{3} = -1$$

$$\sigma_2^2 = \left(\frac{(-1)^2 + 2^2 + (-4)^2}{3} \right) - (-1)^2 = 6$$

Covariance Matrix ($\mathbf{C}$)

$$\mathbf{X}_1 = \{1, 0, 3\} \quad \mathbf{X}_2 = \{-1, 2, -4\}$$

$$\text{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \left(\frac{(-1 \times 1) + (2 \times 0) + (-4 \times 3)}{3} \right) - (-1 \times 1.33) = -3$$

$$\mathbf{C} = \begin{bmatrix} \sigma_1^2 & \text{Cov}(\mathbf{X}_1, \mathbf{X}_2) \\ \text{Cov}(\mathbf{X}_1, \mathbf{X}_2) & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} 2.56 & -3 \\ -3 & 6 \end{bmatrix}$$

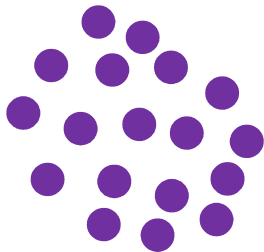
Point Sets in $\mathbb{R}^1$ and $\mathbb{R}^n$



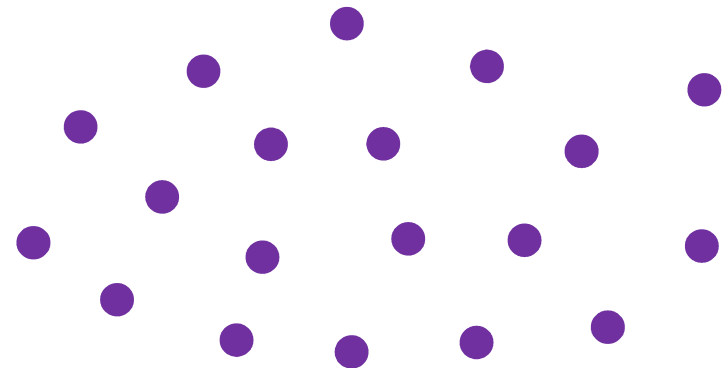
More Compact Point Set in $\mathbb{R}^1$
Low Variance σ^2



More Spread Out Point Set in $\mathbb{R}^1$
High Variance σ^2



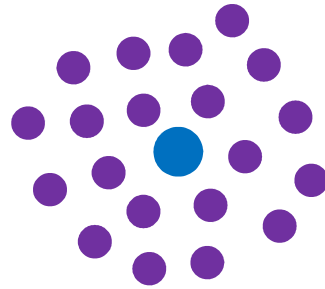
More Compact Point Set in $\mathbb{R}^n$
Low $|\mathcal{C}|$



More Spread Out Point Set in $\mathbb{R}^n$
High $|\mathcal{C}|$

Mahalanobis Distance

x ●



$S \equiv (\mu, C)$

$$d^2(x, S) = (x - \mu)^T C^{-1} (x - \mu)$$

Mahalanobis Distance

$$\mathbf{x} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\mathbf{S} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \end{bmatrix} \right\}$$

$$\boldsymbol{\mu} = \begin{bmatrix} 1.33 \\ -1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 2.56 & -3 \\ -3 & 6 \end{bmatrix}$$

$$d^2(\mathbf{x}, \mathbf{S}) = \left(\begin{bmatrix} -2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1.33 \\ -1 \end{bmatrix} \right)^T \begin{bmatrix} 2.56 & -3 \\ -3 & 6 \end{bmatrix}^{-1} \left(\begin{bmatrix} -2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1.33 \\ -1 \end{bmatrix} \right)$$

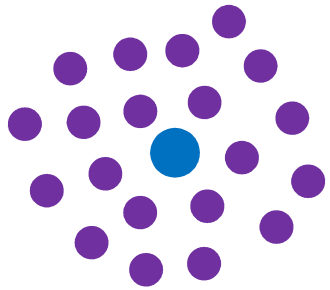
$$d^2(\mathbf{x}, \mathbf{S}) = 4.3354$$



$$d(\mathbf{x}, \mathbf{S}) = 2.0822$$

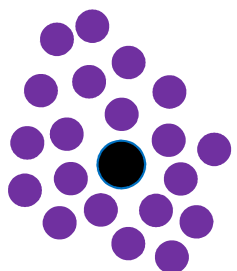
Mahalanobis Distance

$$\mathbf{S}_2 \equiv (\boldsymbol{\mu}_2, \mathbf{C}_2)$$



$$d^2(\boldsymbol{\mu}_2, \mathbf{S}_1) = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \mathbf{C}_1^{-1} (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$$

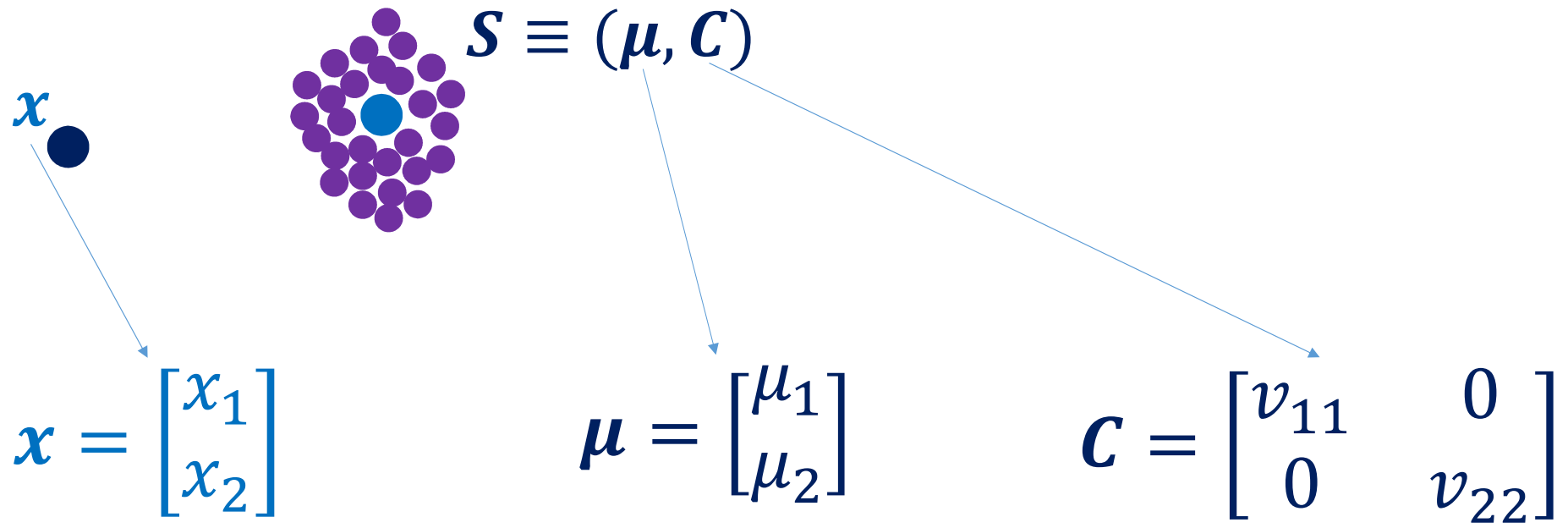
$$d^2(\mathbf{S}_1, \mathbf{S}_2) = \frac{1}{2} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T (\mathbf{C}_1^{-1} + \mathbf{C}_2^{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$



$$\mathbf{S}_1 \equiv (\boldsymbol{\mu}_1, \mathbf{C}_1)$$

$$d^2(\boldsymbol{\mu}_1, \mathbf{S}_2) = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathbf{C}_2^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

Mahalanobis Distance



$$d^2(x, S) = (x - \mu)^T C^{-1} (x - \mu) = ?$$

Mahalanobis Distance

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



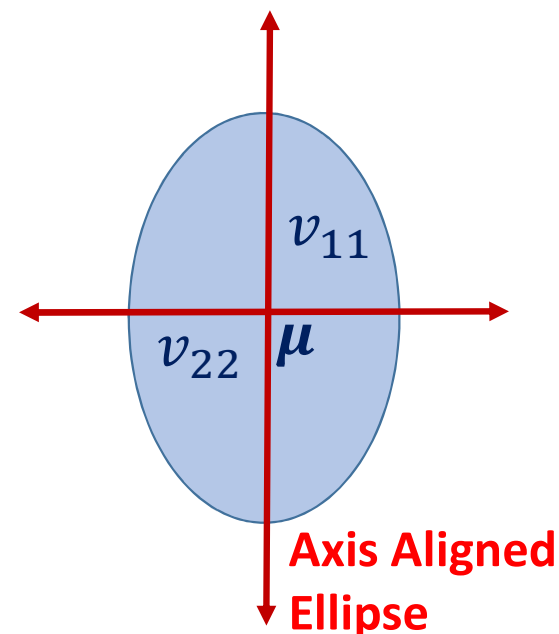
$$\mathbf{S} \equiv (\boldsymbol{\mu}, \mathbf{C})$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

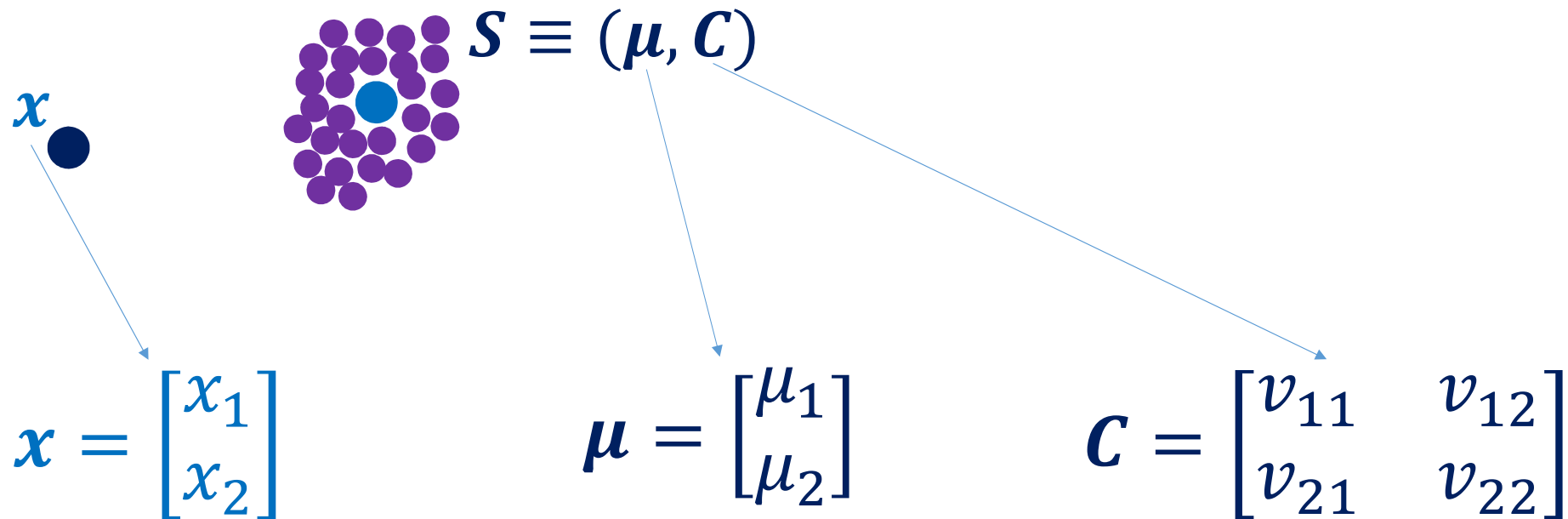
$$\mathbf{C} = \begin{bmatrix} v_{11} & 0 \\ 0 & v_{22} \end{bmatrix}$$

$$d^2(\mathbf{x}, \mathbf{S}) = (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

$$d^2(\mathbf{x}, \mathbf{S}) = \frac{(x_1 - \mu_1)^2}{v_{11}} + \frac{(x_2 - \mu_2)^2}{v_{22}}$$



Mahalanobis Distance

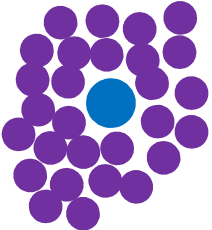


$$d^2(\mathbf{x}, \mathbf{S}) = (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = ?$$

$$v_{12} = v_{21} = v$$

Mahalanobis Distance

$\bullet$
 $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

 $\mathcal{S} \equiv (\boldsymbol{\mu}, \mathbf{C})$

$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$

$\mathbf{C} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$

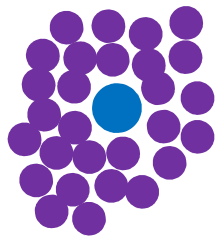
$$\mathbf{C}^{-1} = \frac{1}{v_{22}v_{21} - v^2} \begin{bmatrix} v_{22} & -v \\ -v & v_{11} \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$d^2(\mathbf{x}, \mathcal{S}) = (\mathbf{x} - \boldsymbol{\mu})^T \begin{bmatrix} a & b \\ b & c \end{bmatrix} (\mathbf{x} - \boldsymbol{\mu}) = ?$$

$$v_{12} = v_{21} = v$$

Mahalanobis Distance

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

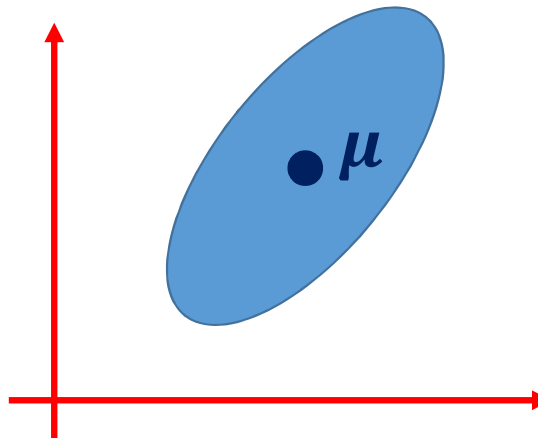


$$\mathcal{S} \equiv (\boldsymbol{\mu}, \mathbf{C})$$

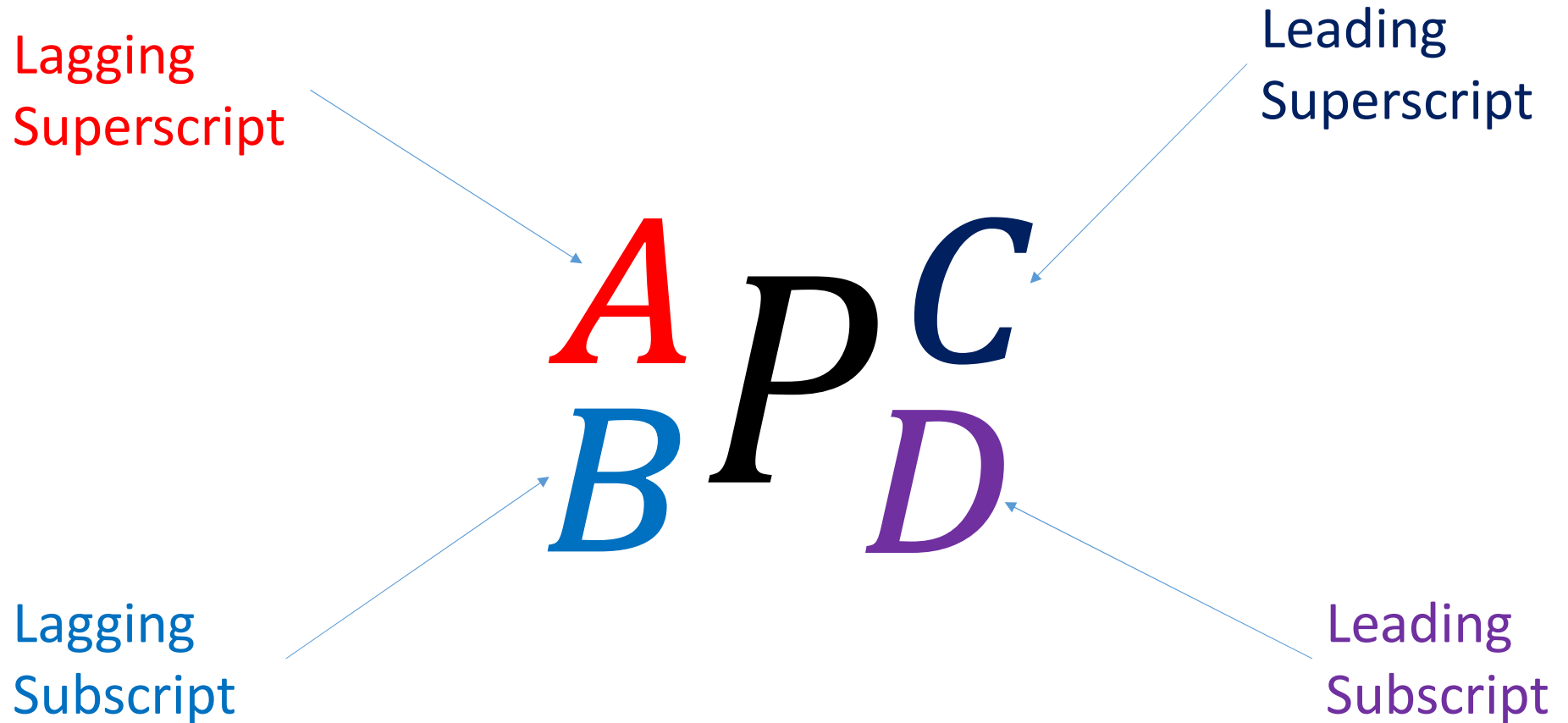
$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$$

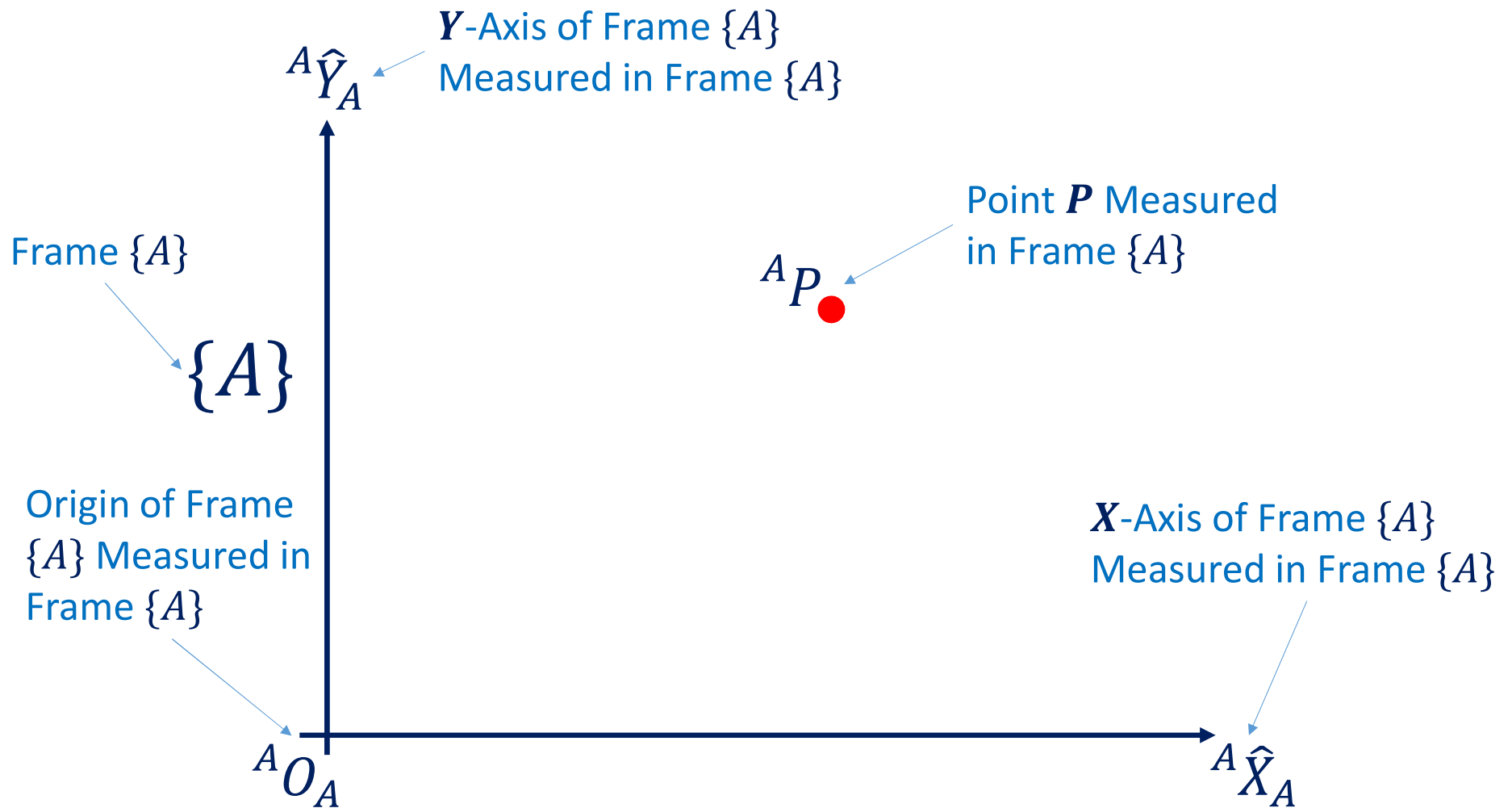
$$d^2(\mathbf{x}, \mathcal{S}) = a(x_1 - \mu_1)^2 + 2b(x_1 - \mu_1)(x_2 - \mu_2) + c(x_2 - \mu_2)^2$$



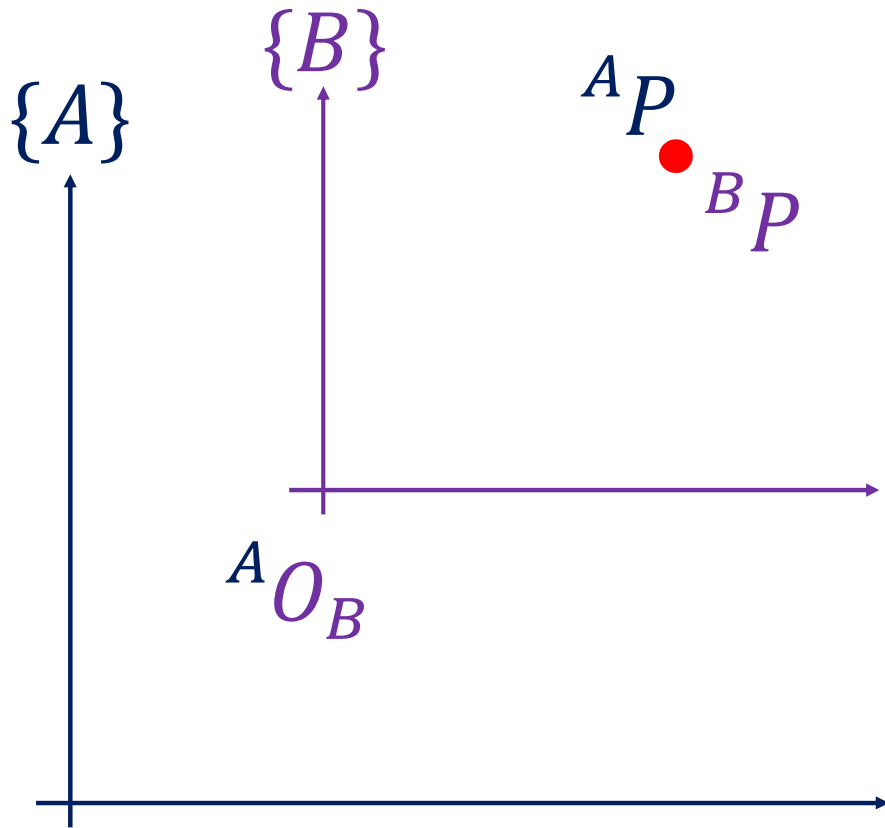
Coordinate Transformations: Notations



Coordinate Transformations: Notations



Coordinate Transformations: Translation

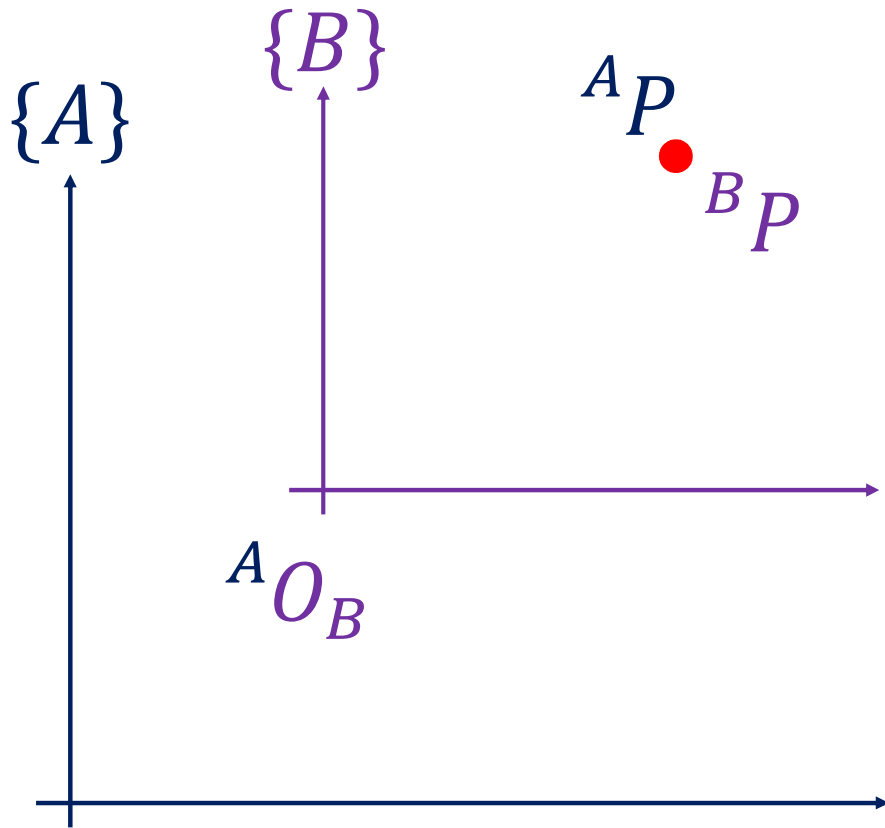


$${}^A \mathbf{P} = (10, 10)$$

$${}^A \mathbf{O}_B = (3, 5)$$

$${}^B \mathbf{P} = ?$$

Coordinate Transformations: Translation



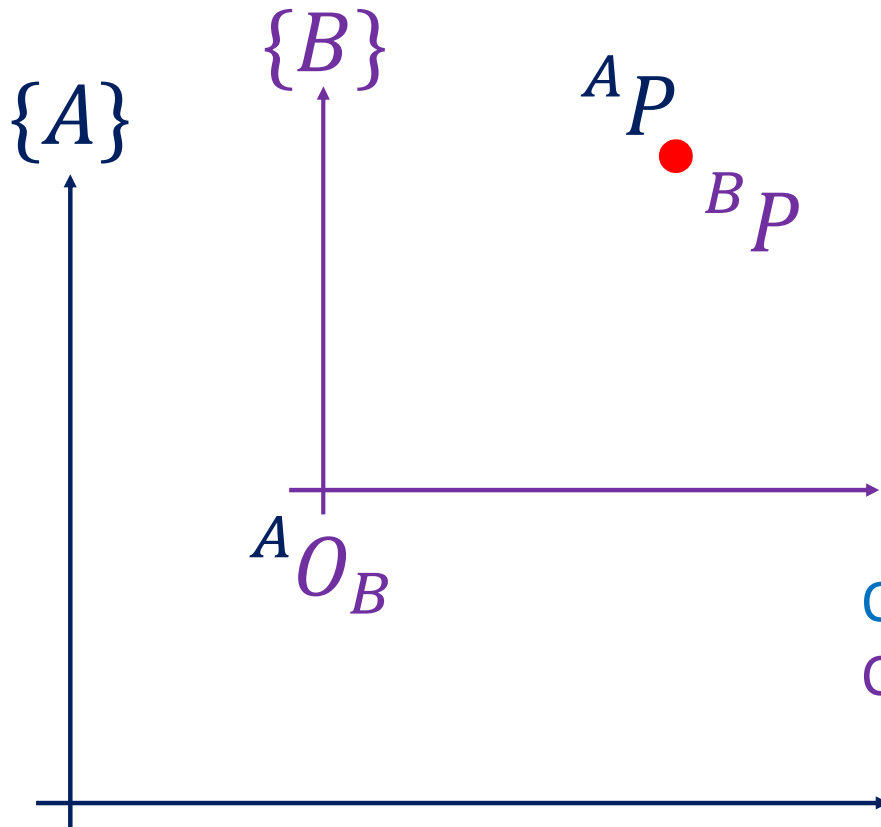
$${}^A \mathbf{P} = (10, 10)$$

$${}^A \mathbf{O}_B = (3, 5)$$

$${}^B \mathbf{P} = (10, 10) - (3, 5)$$

$${}^B \mathbf{P} = (7, 5)$$

Coordinate Transformations: Translation

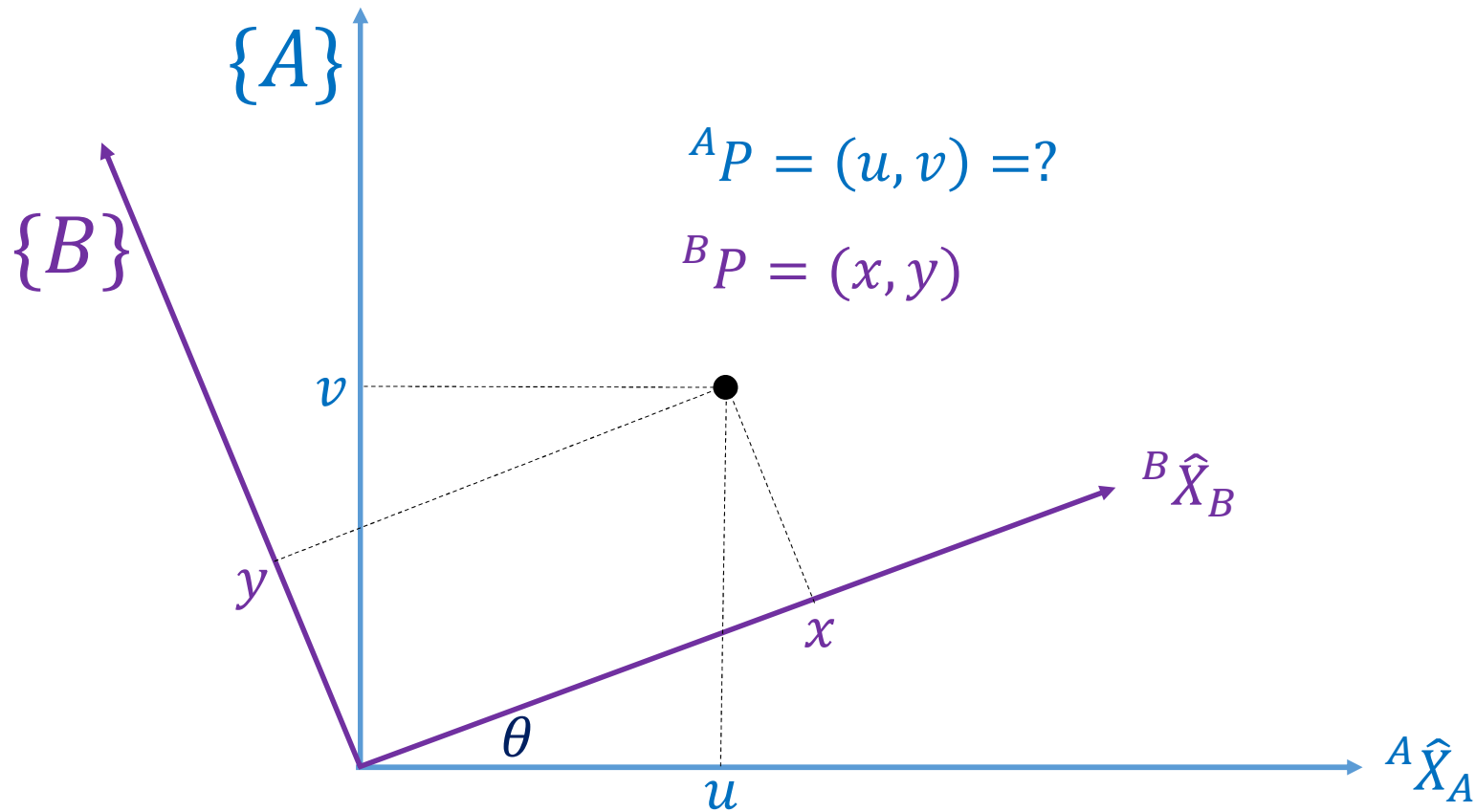


$${}^B \mathbf{P} = {}^A \mathbf{P} - {}^A \mathbf{O}_B$$

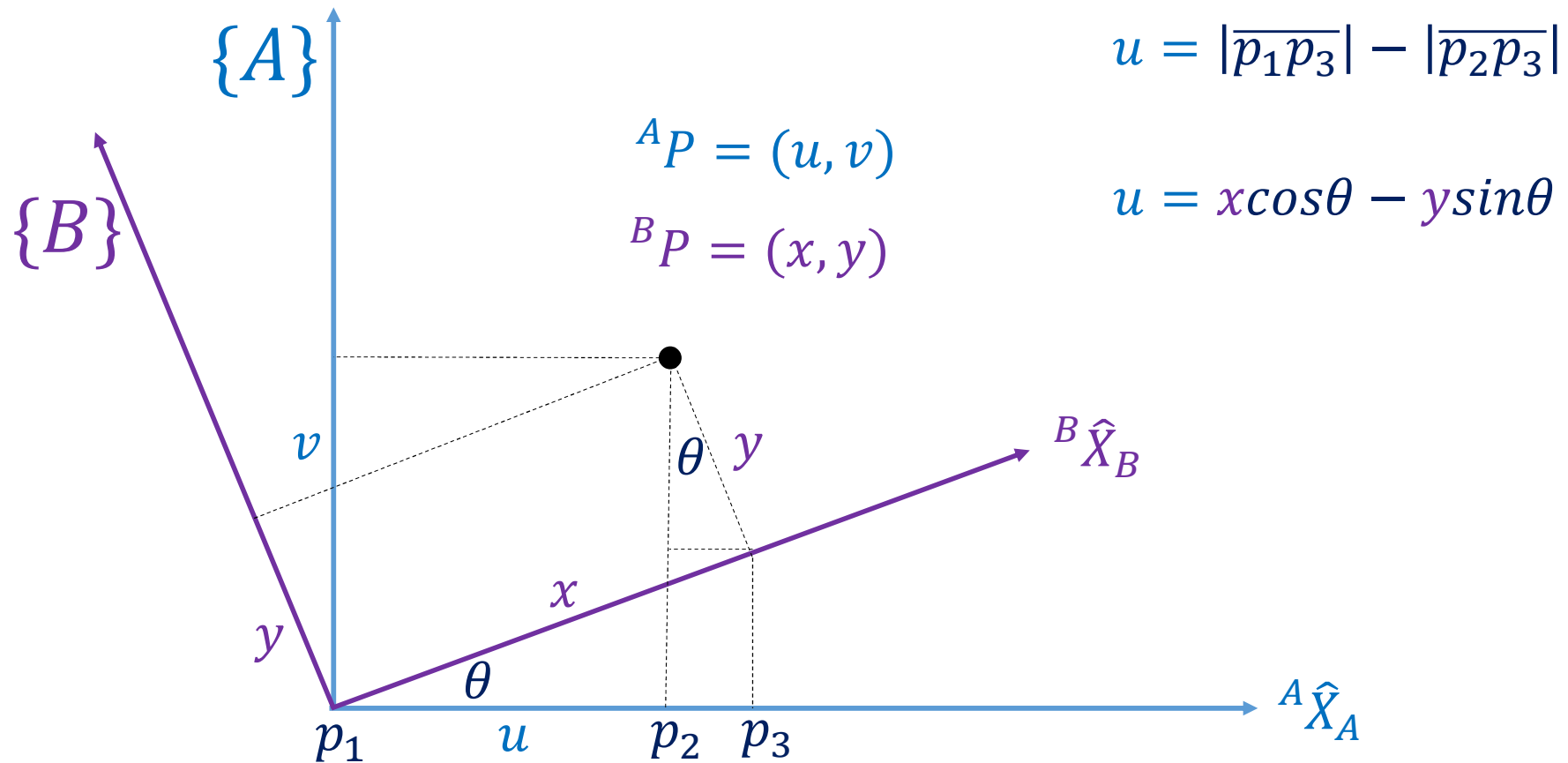
Translation

Origins of $\{A\}$ and $\{B\}$ do not coincide
Corresponding Axes of $\{A\}$ and $\{B\}$ are parallel

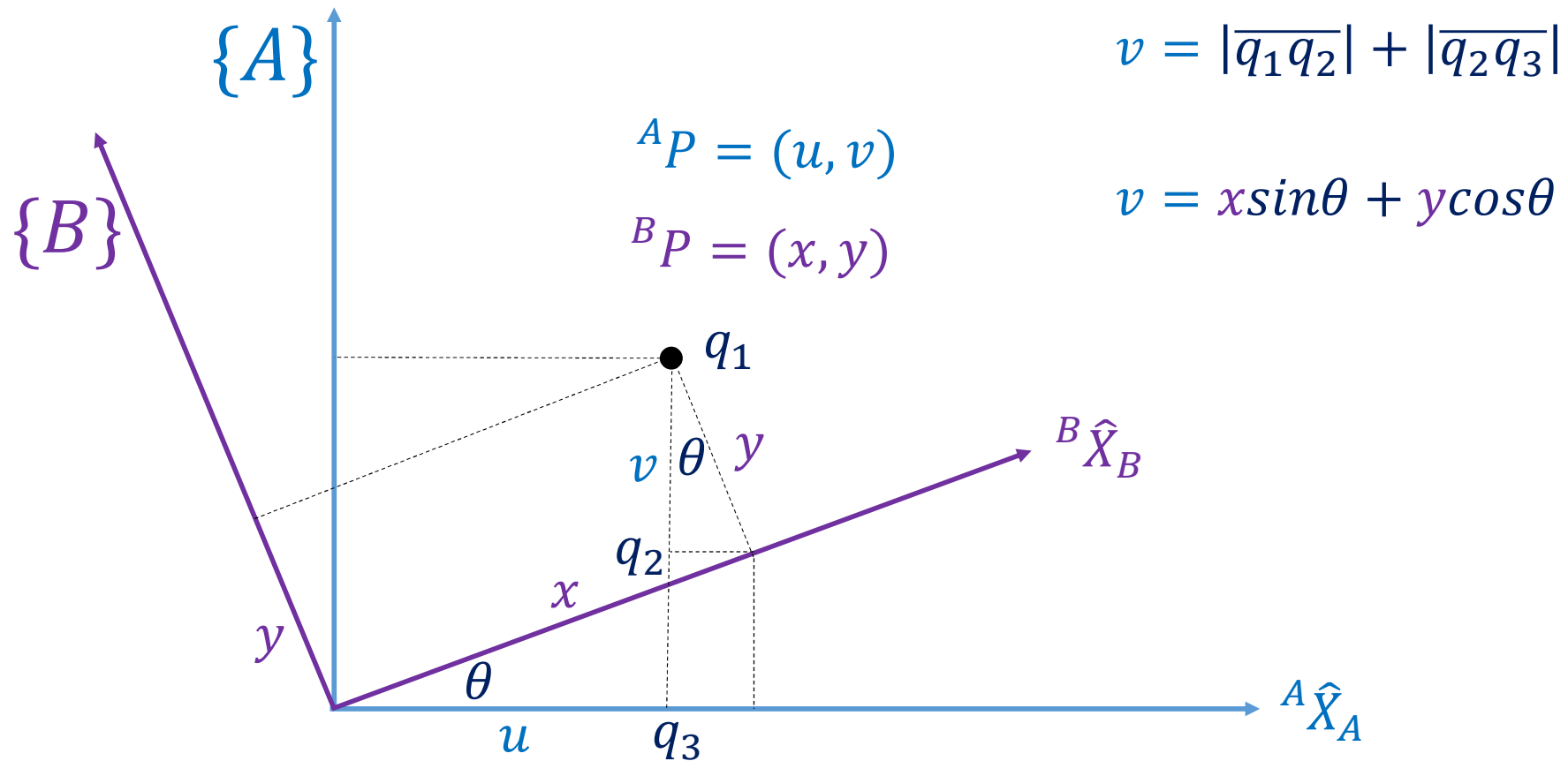
Coordinate Transformations: Rotation



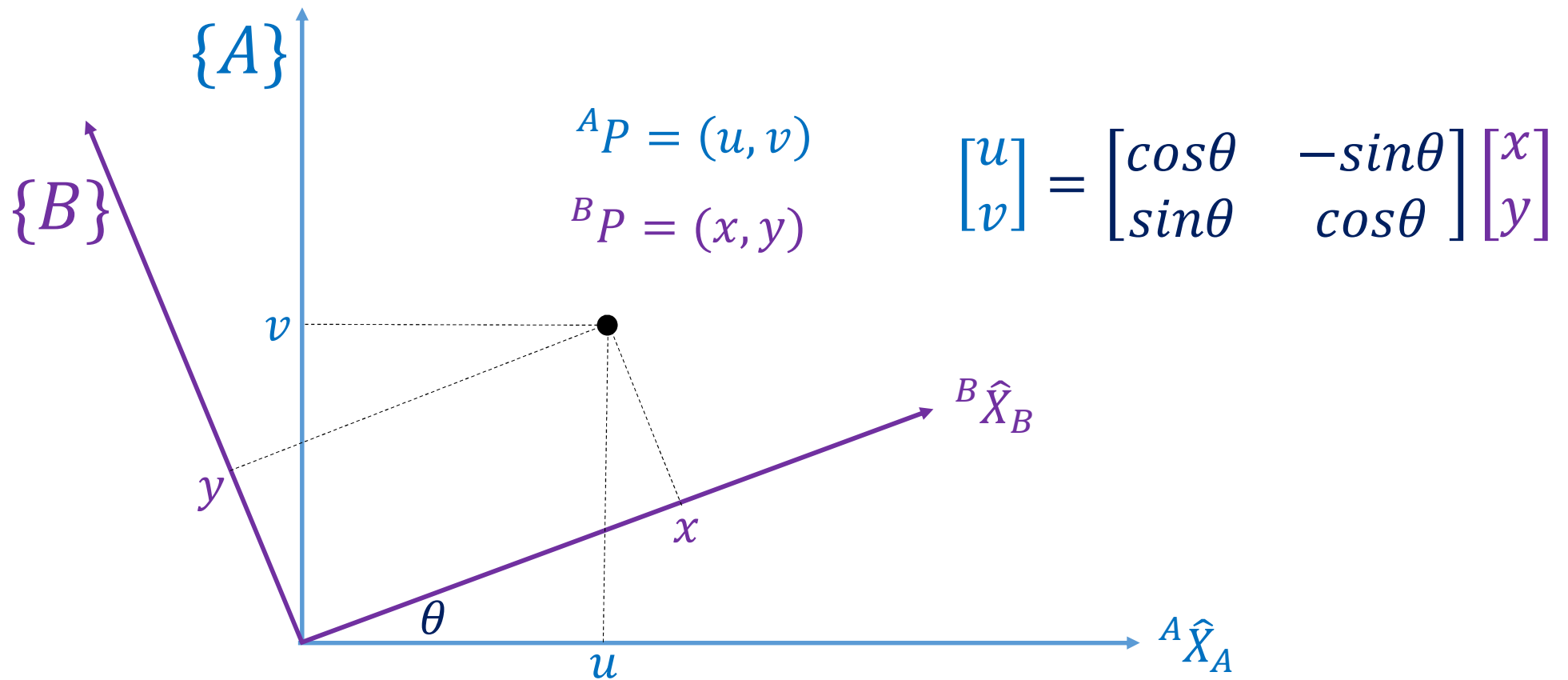
Coordinate Transformations: Rotation



Coordinate Transformations: Rotation



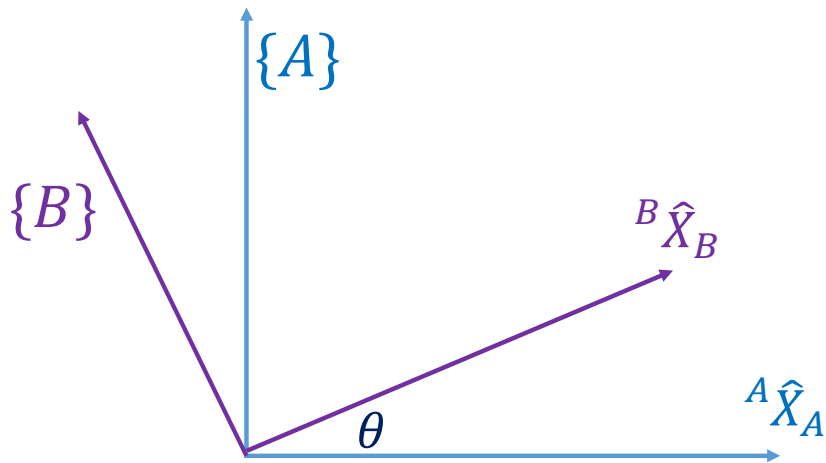
Coordinate Transformations: Rotation



Coordinate Transformations: Rotation

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \longrightarrow \quad {}^A P = {}^A_B R {}^B P$$

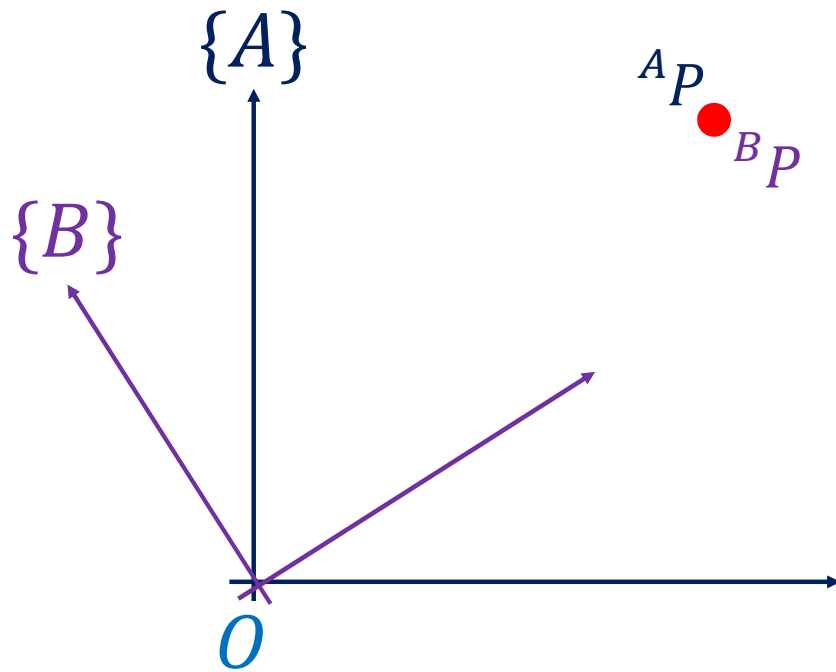
$$|{}^A_B R| = 1 \quad \left\langle \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}, \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \right\rangle = 0 \quad {}^A_B R^T = {}^A_B R^{-1}$$



$${}^A \hat{X}_B = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$${}^A \hat{Y}_B = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

Coordinate Transformations: Rotation



$${}^A P = {}_B^A R {}^B P$$

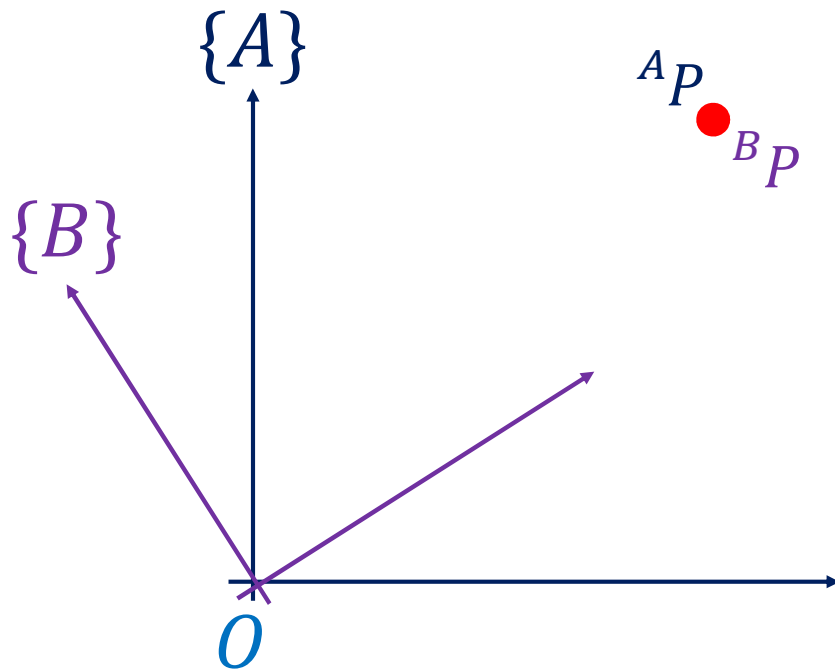
Application of Rotation Matrix ${}_B^A R$
From Reference Frame {B} to Frame {A}

$${}^A \hat{X}_B = {}_B^A R {}^B \hat{X}_B \quad {}^A \hat{Y}_B = {}_B^A R {}^B \hat{Y}_B$$

$$\begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B \end{bmatrix} = {}_B^A R \begin{bmatrix} {}^B \hat{X}_B & {}^B \hat{Y}_B \end{bmatrix}$$

$$\begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B \end{bmatrix} = {}_B^A R \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Coordinate Transformations: Rotation



$${}^A P = {}^A_B R {}^B P$$

Application of Rotation Matrix ${}^A_B R$
From Reference Frame $\{B\}$ to Frame $\{A\}$

$$\begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B \end{bmatrix} = {}^A_B R \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

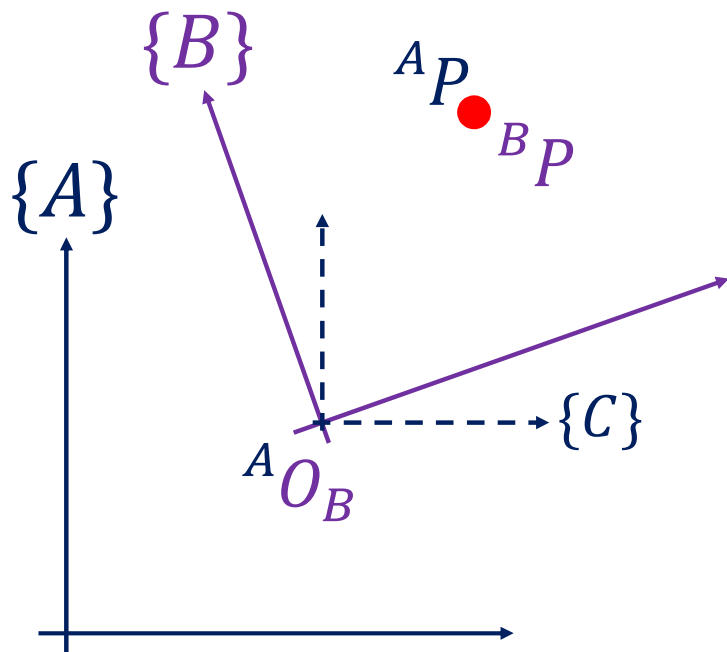


$${}^A_B R = \begin{bmatrix} {}^A \hat{X}_B & {}^A \hat{Y}_B \end{bmatrix}$$

$${}^B_A R = \begin{bmatrix} {}^B \hat{X}_A & {}^B \hat{Y}_A \end{bmatrix}$$

$${}^A P = {}^A_B R {}^B P \quad \longrightarrow \quad {}^B P = {}^A_B R^{-1} {}^A P = {}^A_B R^T {}^A P = {}^B_A R {}^A P$$

Coordinate Transformations: Translation & Rotation



Translate First

$${}^C P = {}^A P - {}^A O_B$$

Rotate Next

$${}^B P = {}^B_C R {}^C P$$

Together

$${}^B P = {}^B_A R ({}^A P - {}^A O_B)$$

Rewritten

$${}^B P = {}^A_B R^T ({}^A P - {}^A O_B)$$

Mahalanobis Distance Revisited

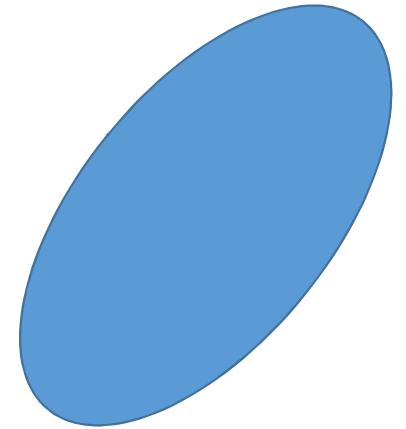
$$(X - \mu)^T C^{-1} (X - \mu) = k \quad \text{C: Non-Diagonal Matrix}$$



Covariance Matrix



Oriented Ellipse



$$C = Q \hat{C} Q^T$$

**Covariance Matrix
Diagonalization**

$$\hat{C} = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}$$

**Diagonal Matrix of
Eigen Values**

$$Q = [\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n]$$

**Orthogonal Matrix of
Eigen Vectors**

Mahalanobis Distance Revisited

$$(X - \mu)^T C^{-1} (X - \mu) = (X - \mu)^T (Q \hat{C} Q^T)^{-1} (X - \mu)$$

$$= (X - \mu)^T (Q^T)^{-1} (\hat{C}^{-1}) (Q)^{-1} (X - \mu)$$

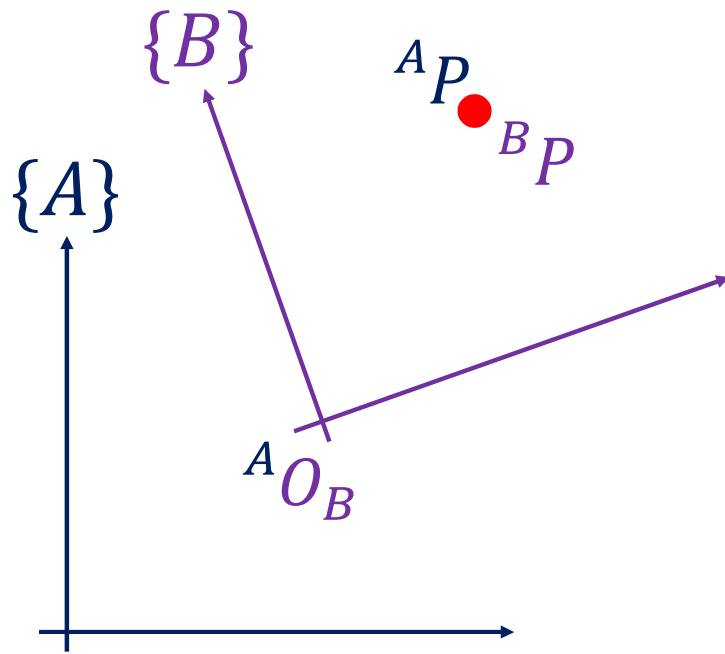
$$= (X - \mu)^T Q (\hat{C}^{-1}) Q^T (X - \mu)$$

$$= (Q^T (X - \mu))^T (\hat{C}^{-1}) Q^T (X - \mu) = U^T \hat{C}^{-1} U$$

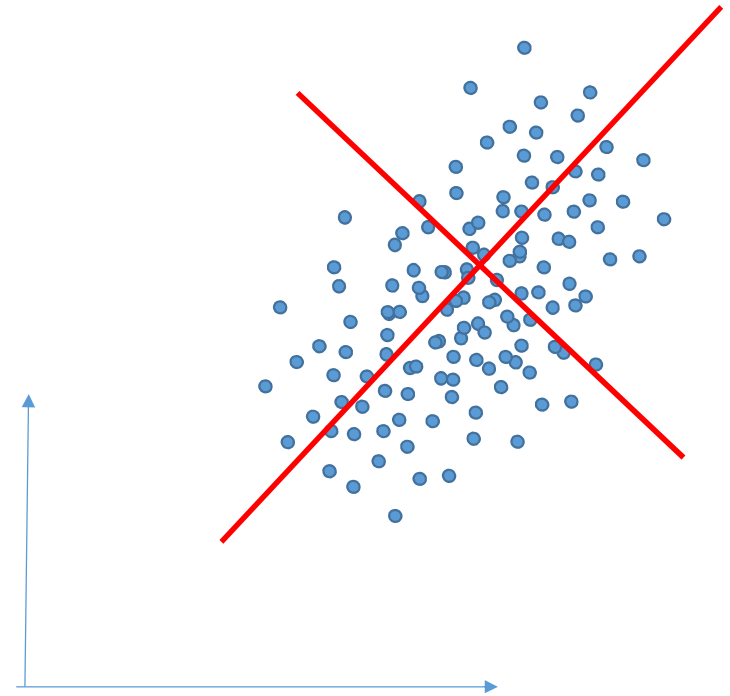
$$= \sum_{i=1}^n \frac{(U[i])^2}{\lambda_i}$$

$$U = Q^T (X - \mu)$$

Relating the Two



$${}^B P = {}^A_B R^T ({}^A P - {}^A O_B)$$



$$U = Q^T (X - \mu)$$

Generation of Rotation Matrix

Generate a Random Rotation Matrix in n dimensional space. How will you do it for $n = 2$?

Generation of Rotation Matrix

Rotation Matrix for $n = 2$

→ Generate Random $\theta \in (0, 2\pi)$

→ Construct $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Generation of Rotation Matrix

Rotation Matrix for n Dimensional Space

- Generate Random Matrix $A \in \mathbb{R}^{n \times n}$
- Construct Symmetric Matrix $B = A + A^T$
- Obtain Eigen Vectors $\{e_1, \dots, e_n\}$ of B
- Construct $R = [e_1 \quad \dots \quad e_n]$

Summary

- Frequency Distribution
- Sum Normalized Frequency Distribution
- Measures of Central Tendency
- Standard Deviation & Variance
- Covariance Matrix
- Mahalanobis Distance
- Coordinate Transformation
- Interpreting Mahalanobis Distance



Thank You